

Yiyun He

University of California San Diego

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Background

- ICERM Postdoctoral Fellow Jan 2026 – May 2026
Institute for Computational and Experimental Research in Mathematics, Brown University
 - Stefan E. Warschawski Visiting Assistant Professor Jul 2025 – Current
University of California San Diego
 - Ph.D. degree in Mathematics Sep 2020 – Jun 2025
M.S. degree in Mathematics Sep 2020 – Jun 2022
University of California Irvine
Advisor: **Prof. Roman Vershynin**
 - B.S. degree in Mathematics and Applied Mathematics Sep 2016 – Jun 2020
University of Science and Technology of China, China
Advisor: **Prof. Xiande Zhang**
Special class of the Gifted Young
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Preprints and Papers

- *Concentration of Bounded Sparse Chaoses and Input-Aware Guarantees for Sparse Sketching*
with Guozheng Dai, Ke Wang, Yizhe Zhu
To appear Fall 2026.
- *Data-Dependent Wasserstein Guarantees for Private Adaptive Partitions*
To appear Fall 2026.
- *Perturbation bounds for low-rank matrix approximation algorithms*
with Alberto Bucci, Christopher Musco, Swati Padmanabhan, David Persson, Arvind Saibaba
To appear Fall 2026.
- *FL-Sailer: Efficient and Privacy-Preserving Federated Learning for Scalable Single-Cell Epigenetic Data Analysis via Adaptive Sampling*
with Guangyi Zhang, Yi Dai, Junhao Liu.
Published at **Transactions on Machine Learning Research (TMLR)**, In Press.
- *Minimax Optimal Differentially Private Synthetic Data for Smooth Queries*
with Rundong Ding, Yizhe Zhu.
Extended abstract accepted at **Conference on Learning Theory (COLT)**, 2026.
Submitted to **IEEE, Transactions on Information Theory**.
- *A Note on the Improved Sparse Hanson-Wright Inequalities*
with Guozheng Dai, Ke Wang, Yizhe Zhu.
Preprint, 2025
- *Sparse Hanson-Wright Inequalities with Applications*
with Ke Wang, Yizhe Zhu.
Accepted at **Electronic Journal of Probability**, 2026
- *Online Differentially Private Synthetic Data Generation*
with Roman Vershynin, Yizhe Zhu.
Published at **IEEE Transactions on Privacy**, 2024
- *Differentially private low-dimensional representation of high-dimensional data*

with Thomas Strohmer, Roman Vershynin, Yizhe Zhu.

Published at **Information and Inference**, 2024

- *Algorithmically Effective Differentially Private Synthetic Data*

with Roman Vershynin, Yizhe Zhu.

Published at **Conference on Learning Theory (COLT)**, 2023

- *Strong quantum nonlocality and unextendibility without entanglement in N -partite systems with odd N*

with Fei Shi, Xiande Zhang.

Published at **Quantum**, 2023

Invited and Contributed Talks

- Computational and Applied Mathematics Seminar, NC State University Sept 2026
 - Probability Seminar, Fields Institute, University of Toronto May 2026
 - Probability Seminar, Brown University Apr 2026
 - ICERM Graduate Seminar, ICERM, Brown University Feb 2026
 - Probability Seminar, University of California, San Diego Jan 2025
 - Joint Mathematics Meetings, Seattle, WA Jan 2025
 - Probability Seminar, University of Southern California Aug 2024
 - Group Seminar of Prof. Deanna Needell, University of California, Los Angeles May 2024
 - Southern California Applied Mathematics Symposium Apr 2024
 - Probability Seminar, University of Washington Apr 2024
 - Joint Mathematics Meetings, San Francisco, CA Jan 2024
 - Conference on Learning Theory Jul 2023
 - Southern California Applied Mathematics Symposium Apr 2023
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Teaching Experience

- Instructor, *MATH 11: Introductory Probability & Statistics, lower division*, UCSD Fall 2026
 - Instructor, *MATH 10A: Calculus I, lower division*, UCSD Fall 2025
 - Teaching Assistant, *MATH 3A: Calculus I, lower division*, UCI Summer 2024
 - Teaching Assistant, *MATH 130B: Probability II, upper division*, UCI Spring 2024
 - Teaching Assistant, *MATH 121A: Linear Algebra I, upper division*, UCI Fall 2021
 - Reader, *MATH 121A: Linear Algebra I, upper division*, UCI Summer 2021
 - Reader, *MATH 140A: Elementary Analysis I, upper division*, UCI Spring 2021
 - Teaching Assistant, *Combinatorics*, USTC Fall 2019
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Directed Reading Program

- *Differential privacy in machine learning*, UCI Spring 2024
with Jiaxin Sara-Grace Lien, Mitra Tara Rezvany, Alejandro Salmeron-Tarancon, Kaijie Zhang
- *Pattern Recognition and Bayesian Analysis*, UCI Winter 2025
with Haochen Tan, David Wang
- *Random Matrix Theory*, UCI Spring 2025
with Parsa Gorji, Harita Kalvai, Harita Kalvai, Shiya Tang, Vincent Wang,